Tinkering with Bluetooth

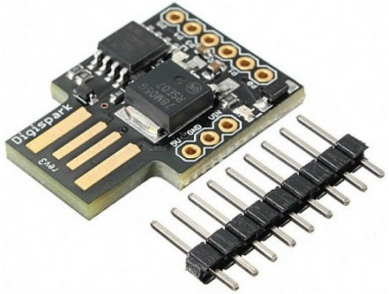


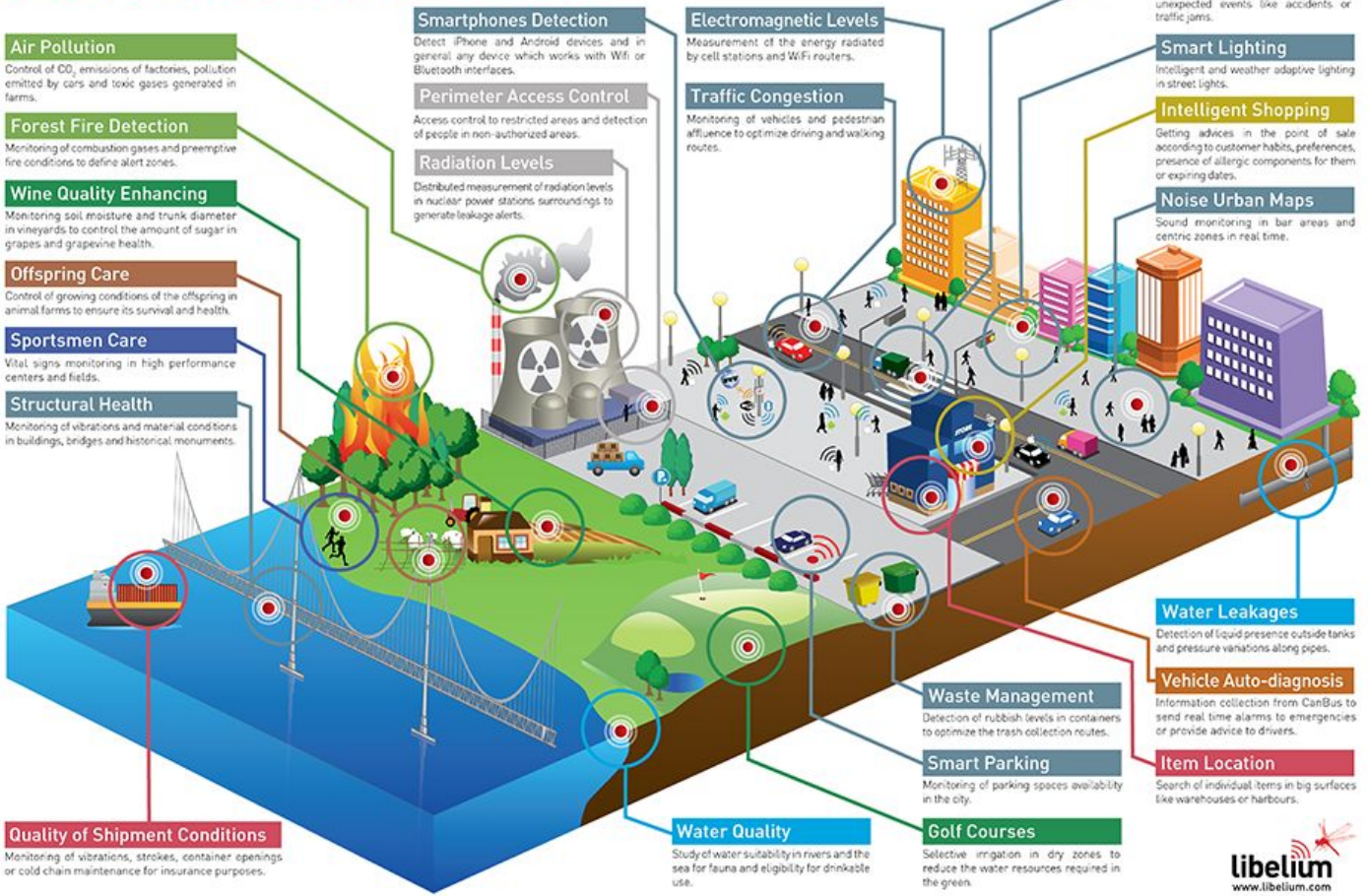
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16/10/2019

The Internet-of-Things *thing*





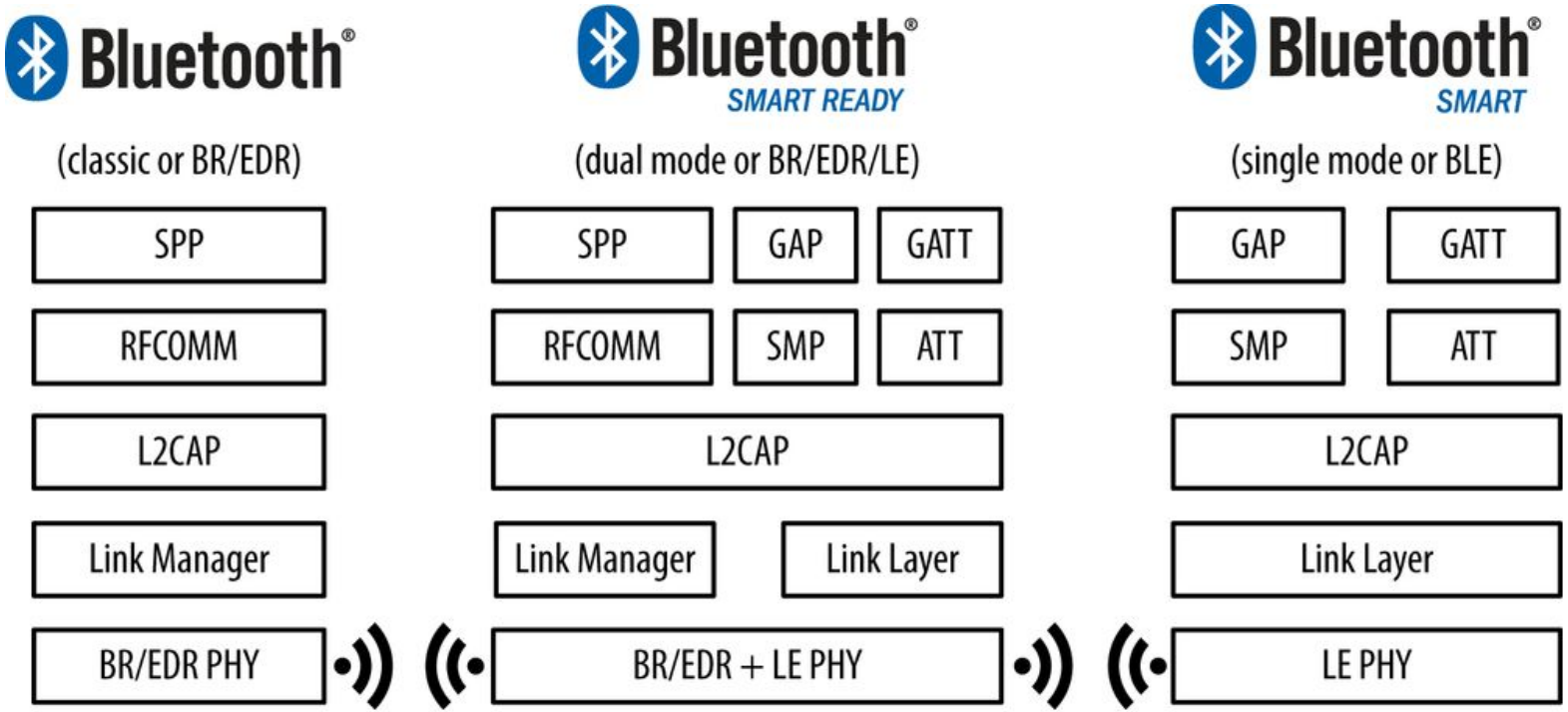


Bluetooth

“Bluetooth is a **wireless** technology standard for exchanging data **between fixed and mobile devices over short distances** using short-wavelength UHF radio waves in the industrial, scientific and medical radio bands, from 2.400 to 2.485 GHz, and building **personal area networks** (PANs).”

From Wikipedia, https://en.wikipedia.org/wiki/Bluetooth

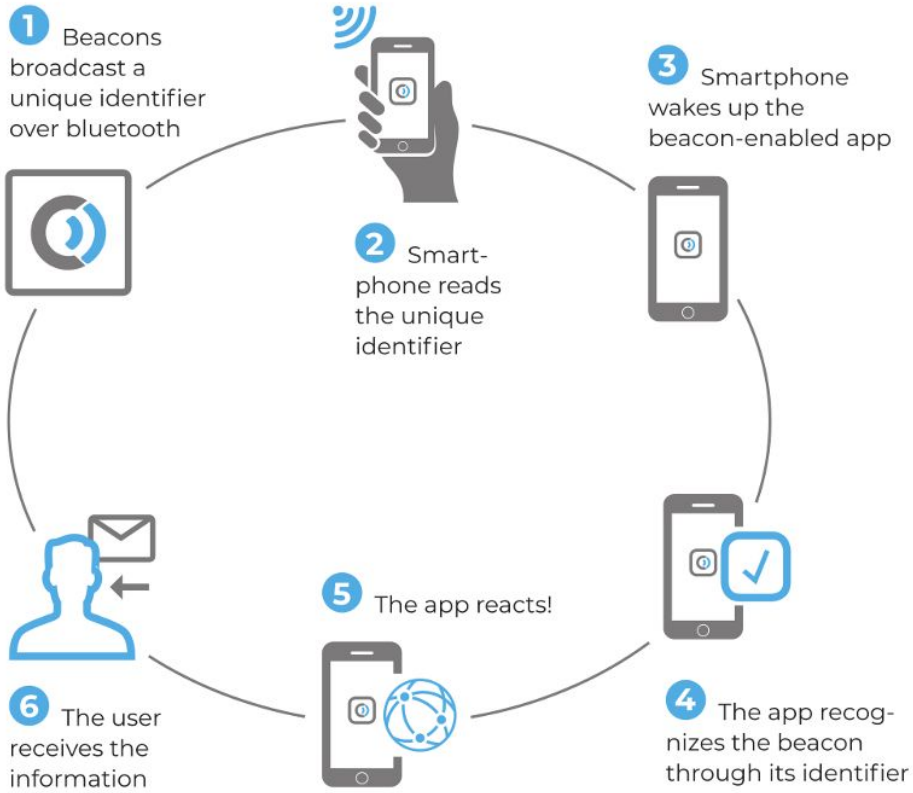
From Bluetooth to BLE



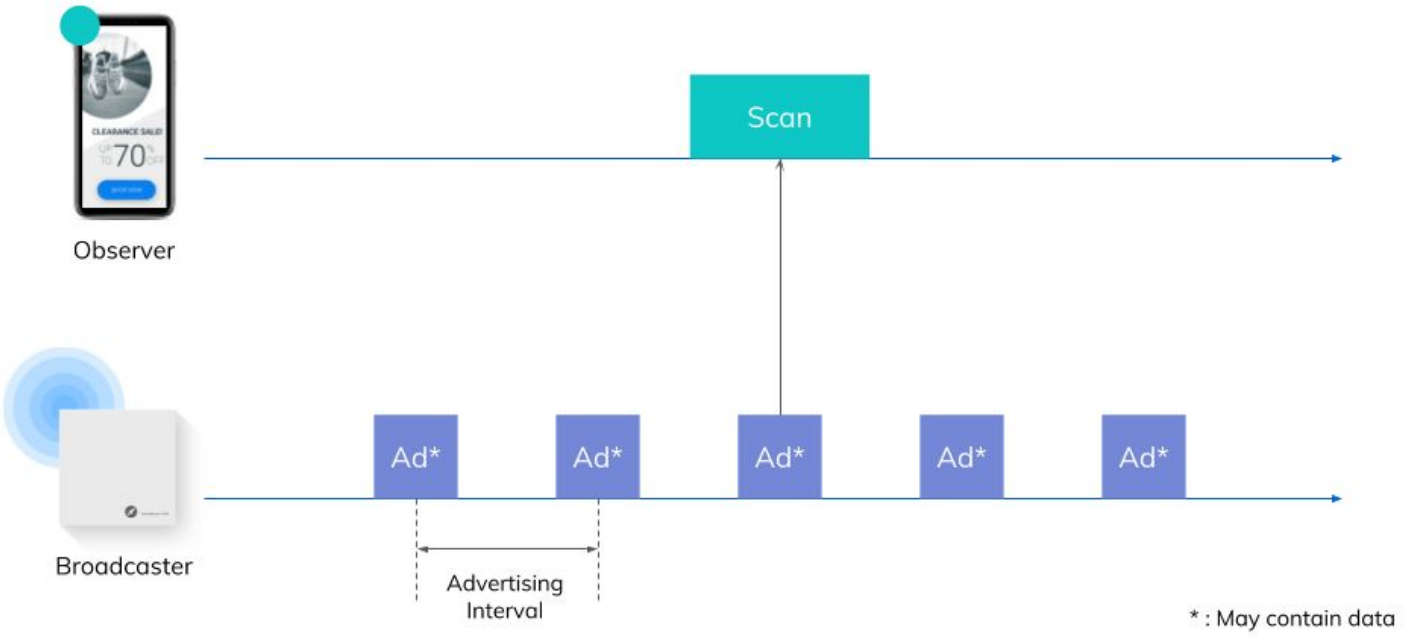
BLE Beacons

"(...) **hardware transmitters** - a class of Bluetooth low energy (LE) **devices that broadcast their identifier to nearby portable electronic devices**"

From Wikipedia, https://en.wikipedia.org/wiki/Bluetooth_low_energy_beacon



How beacons broadcast information?



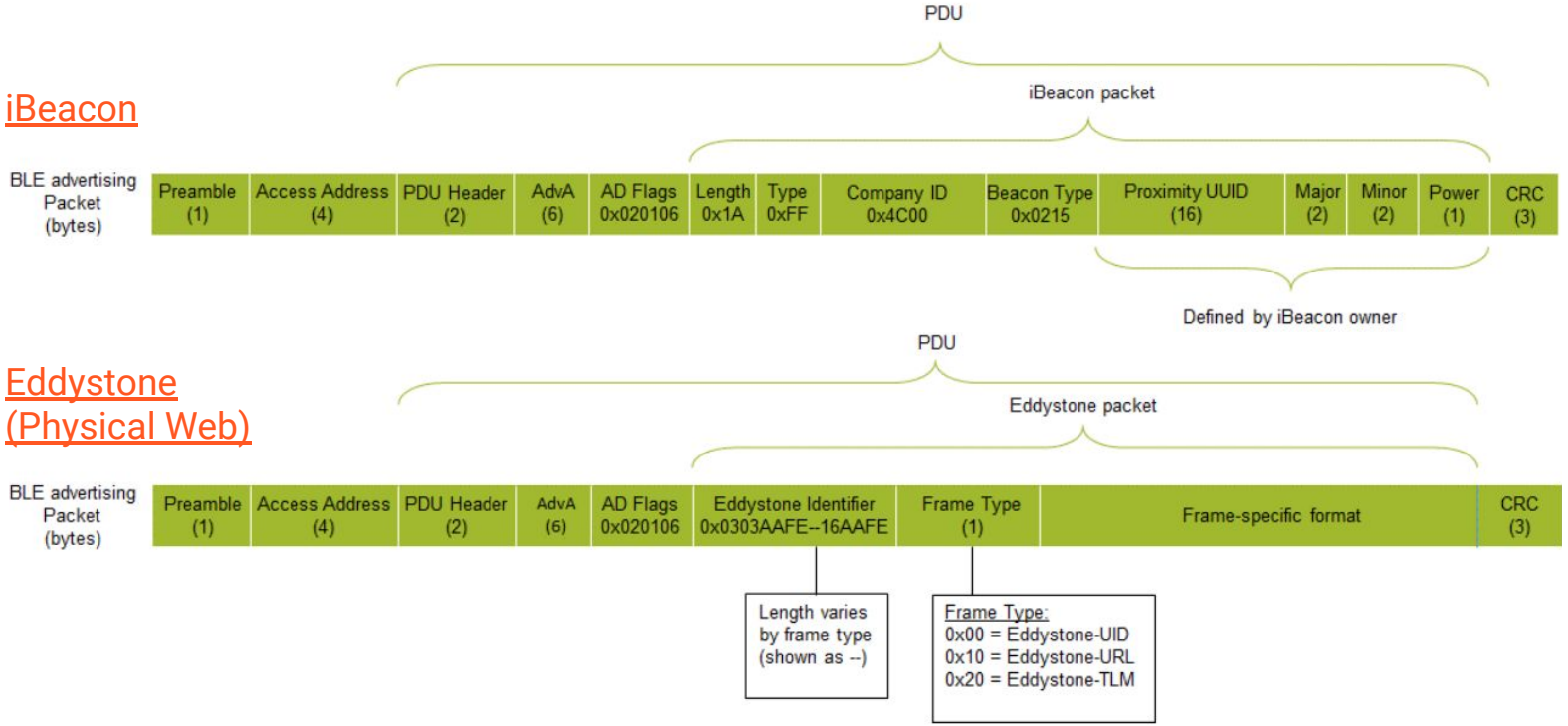
BLE Beacon Protocols



- iBeacon (Apple)
  - Transmits an UUID (string of 24 numbers)
- Eddystone (Google) former UriBeacon
  - Can transmit from one to three frametypes simultaneously:
    - URL (Physical Web, https://google.github.io/physical-web/)
    - UID (similar to Apple's UUID) (string of 16 characters)
    - TLM: **sensor** and **administrative data** from the beacon itself is communicated through telemetry (e.g. battery level, temperature)

- AltBeacon (Radius Networks): Open-source solution.
- GeoBeacon (Tecno-World): Adapted to transmit geographical positions.

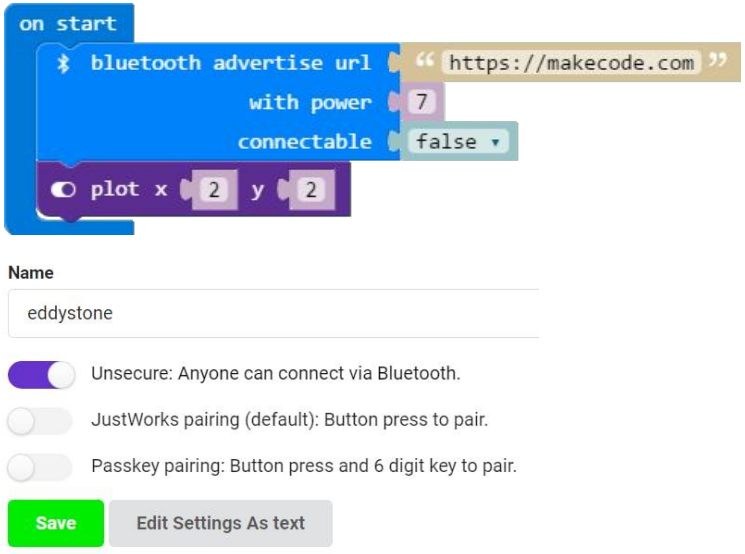
Beacon Frame Structure

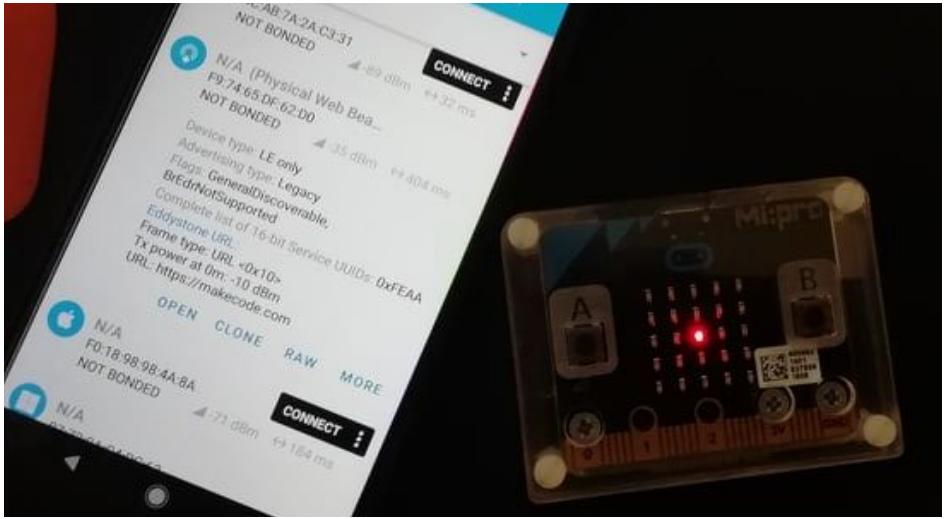


Silicon Labs, Developing Beacons with Bluetooth® Low Energy (BLE) Technology, http://pages.silabs.com/rs/634-SLU-379/images/Whitepaper-Developing-Beacons-with-Bluetooth-Low-Energy-Technology.pdf

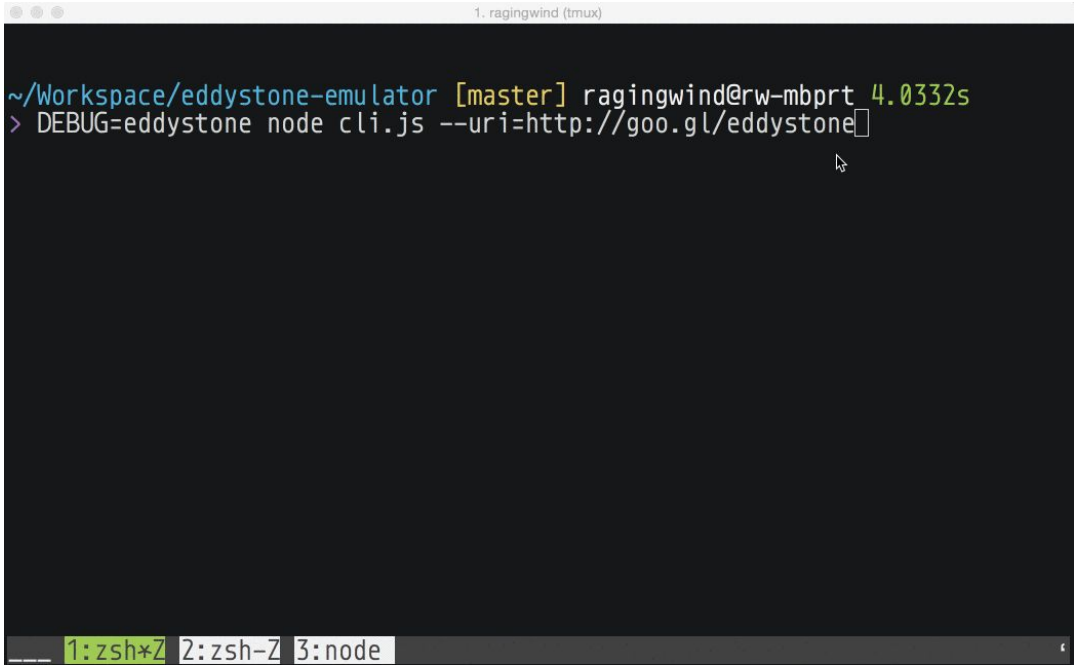
BBC micro:bit beacon

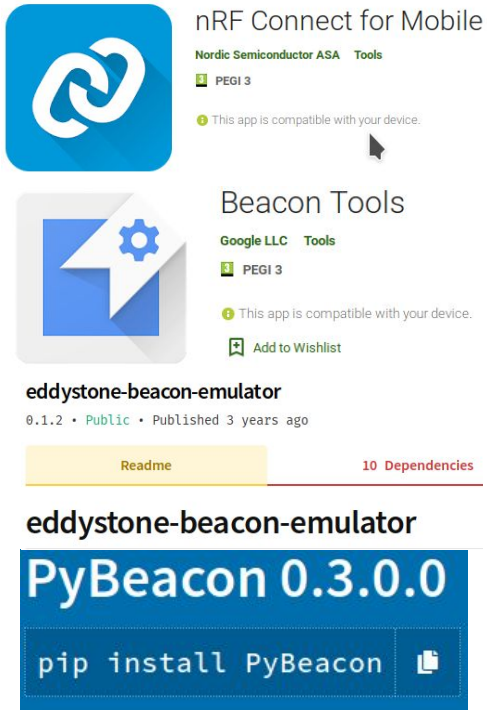
```
bluetooth.advertiseUrl("https://fe.up.pt", 7, false)
led.plot(2, 2)
```





Simulating a Beacon





Exploring the Physical Web (Without Buying Beacons),
https://medium.com/@urish/exploring-the-physical-web-without-buying-beacons-efae51e36c2e

Google Services

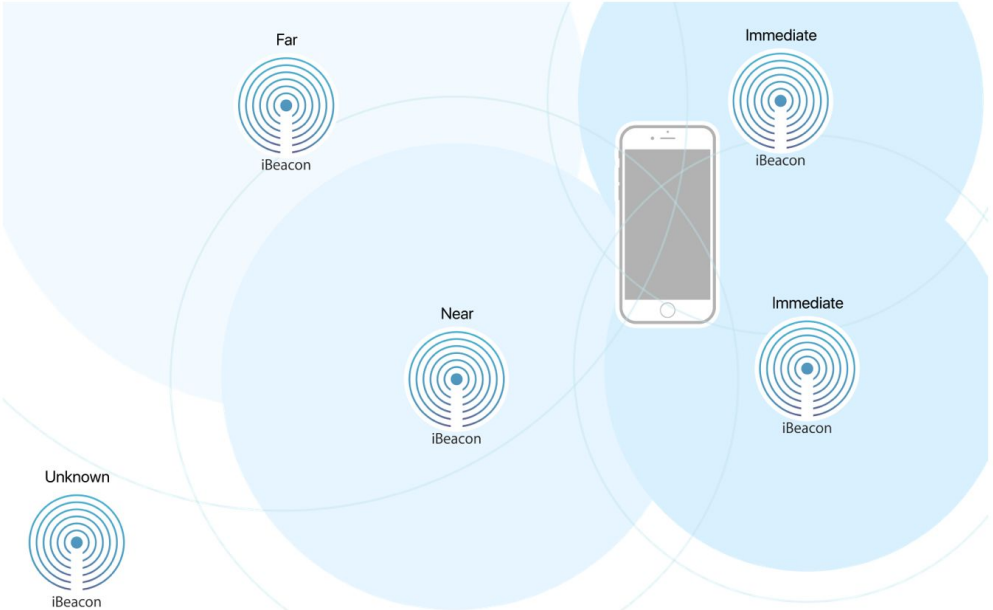


Get Started with Beacons: https://developers.google.com/beacons/get-started
Nearby API: https://developers.google.com/nearby
Proximity Beacon API: https://developers.google.com/beacons/proximity/guides

How to do a BLE localization service?

Open-source projects in the wild:

- https://github.com/neXenio/BLE-Indoor-Positioning
- https://github.com/IndoorLocation/basic-beacon-indoor-location-provider-android





“walk up and use anything”
--The Physical Web

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